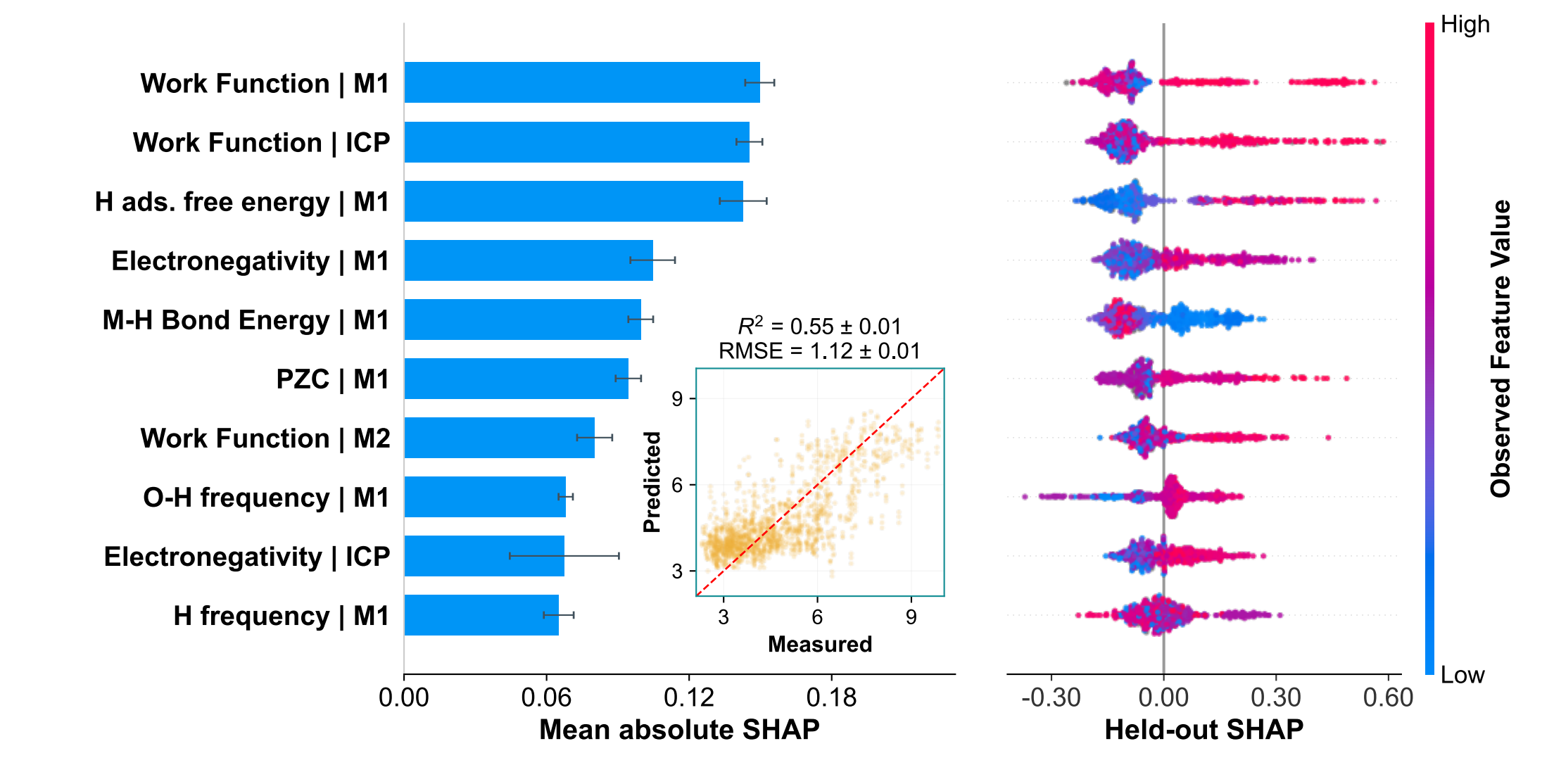
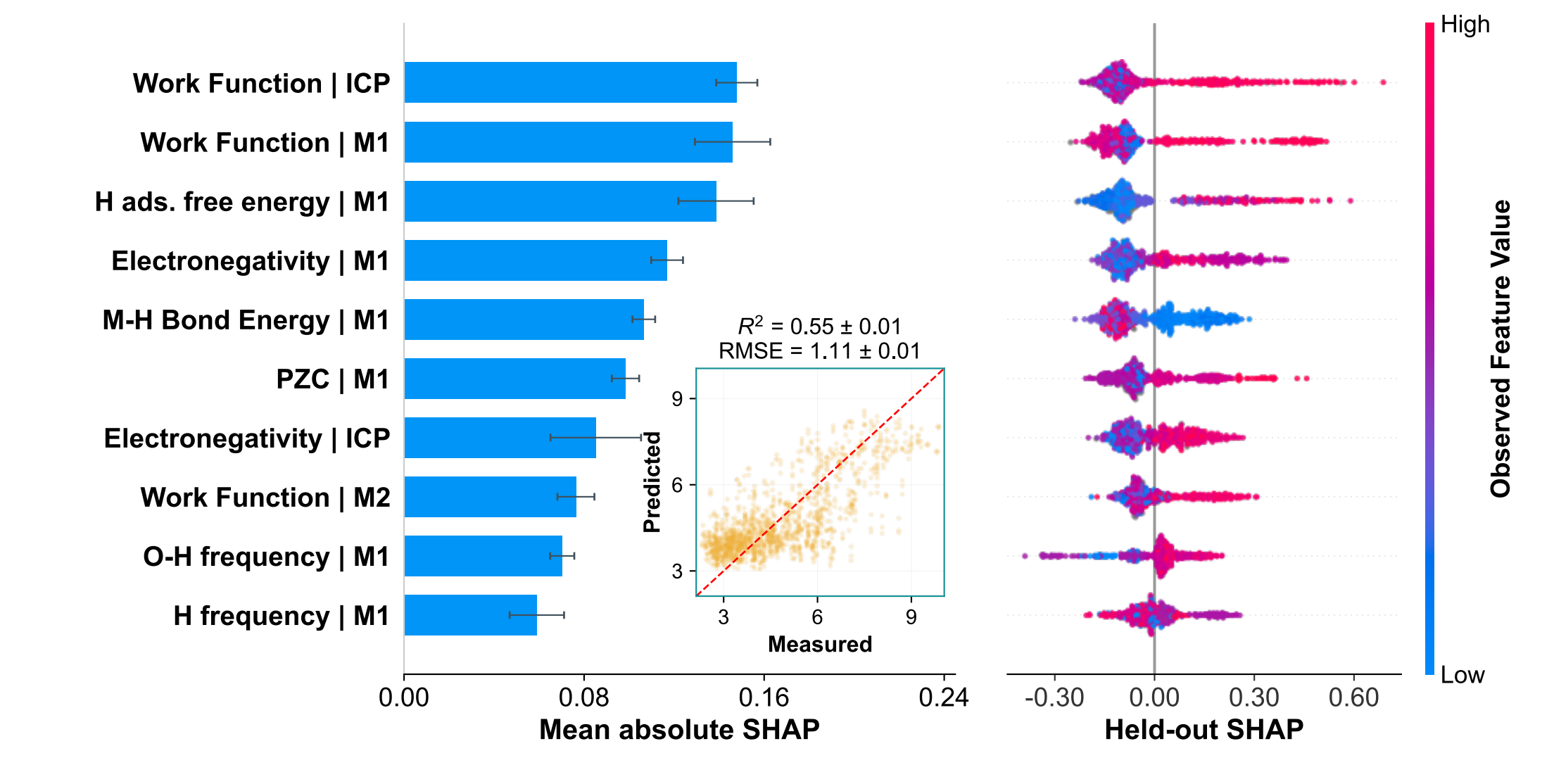
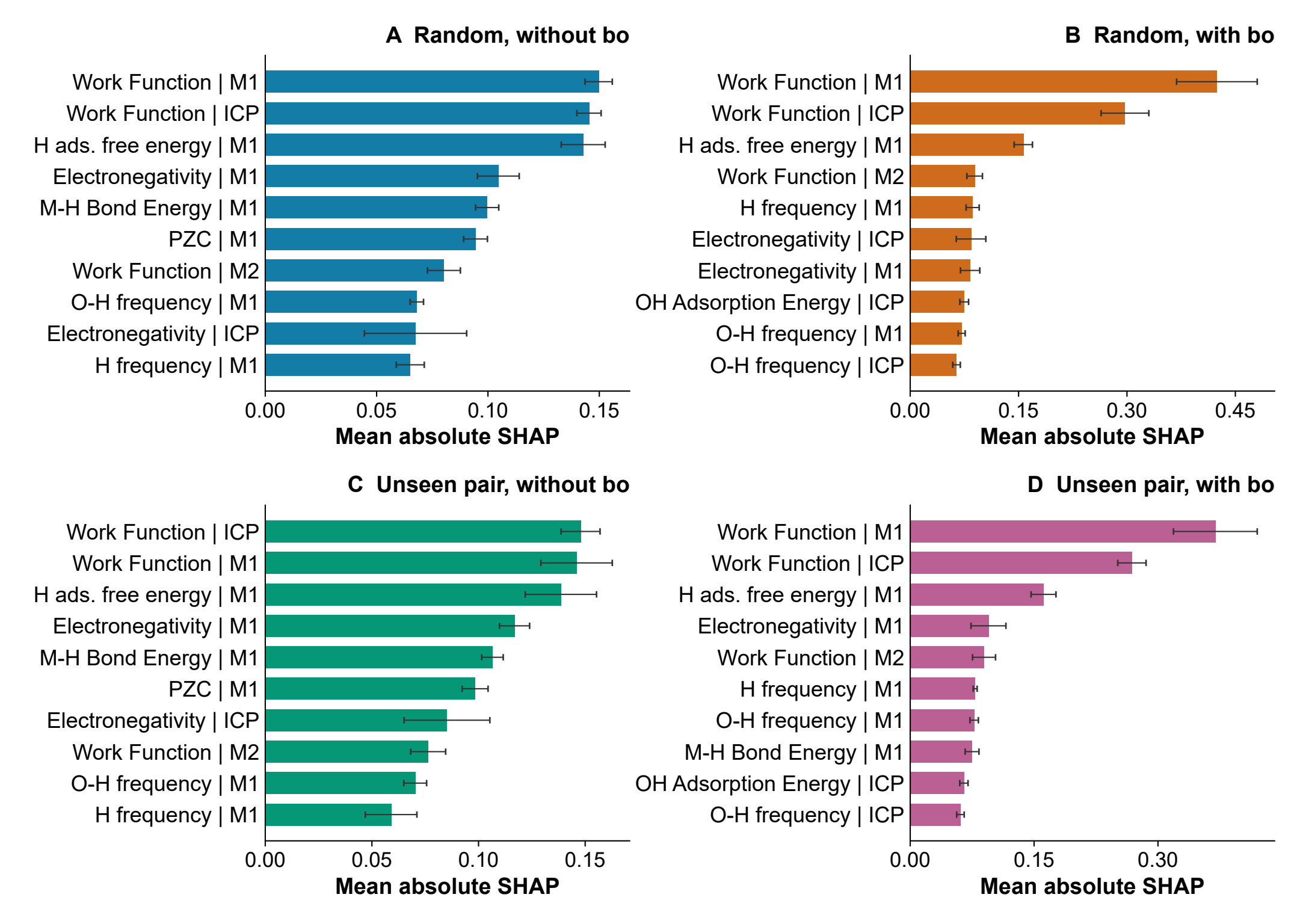
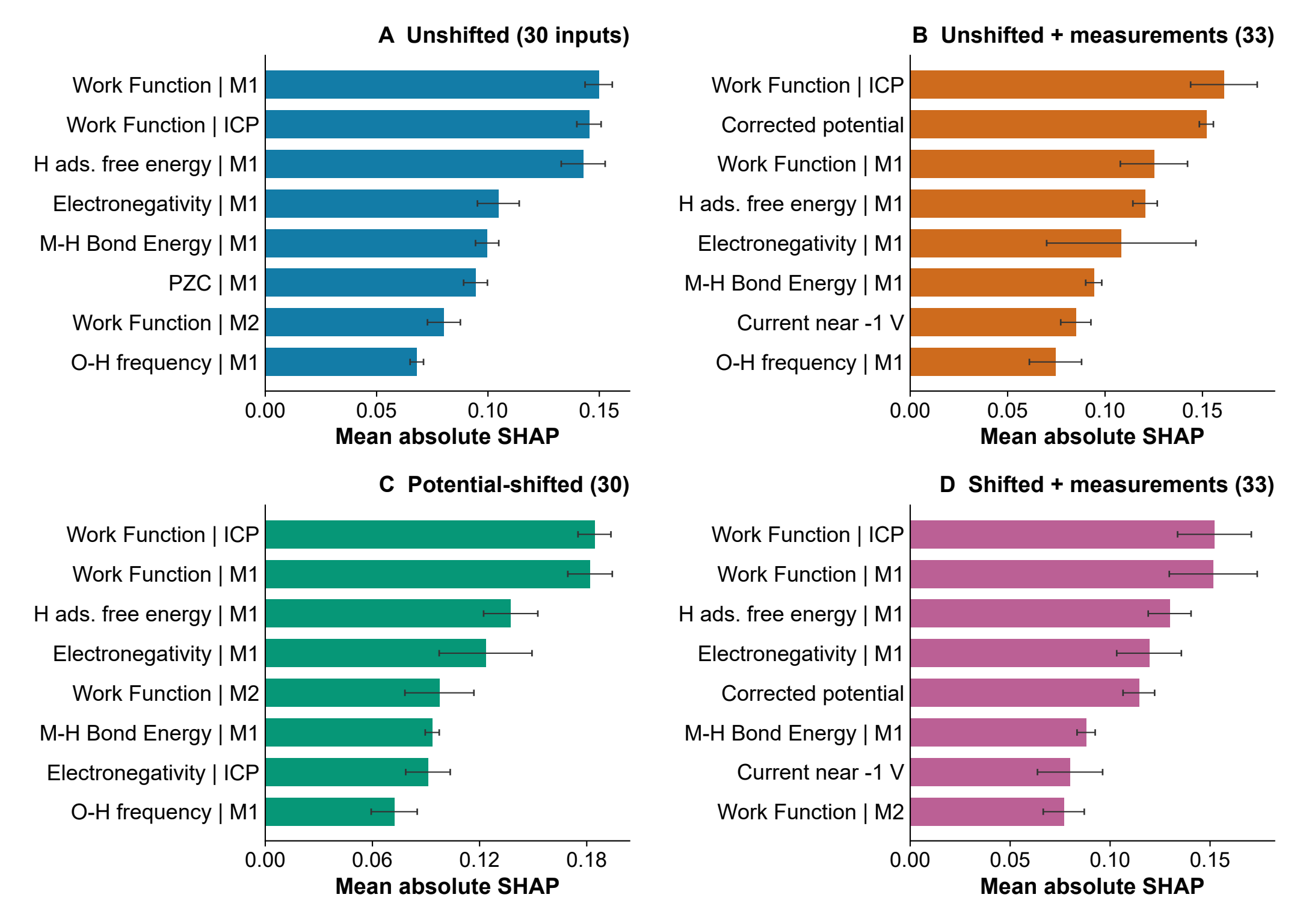
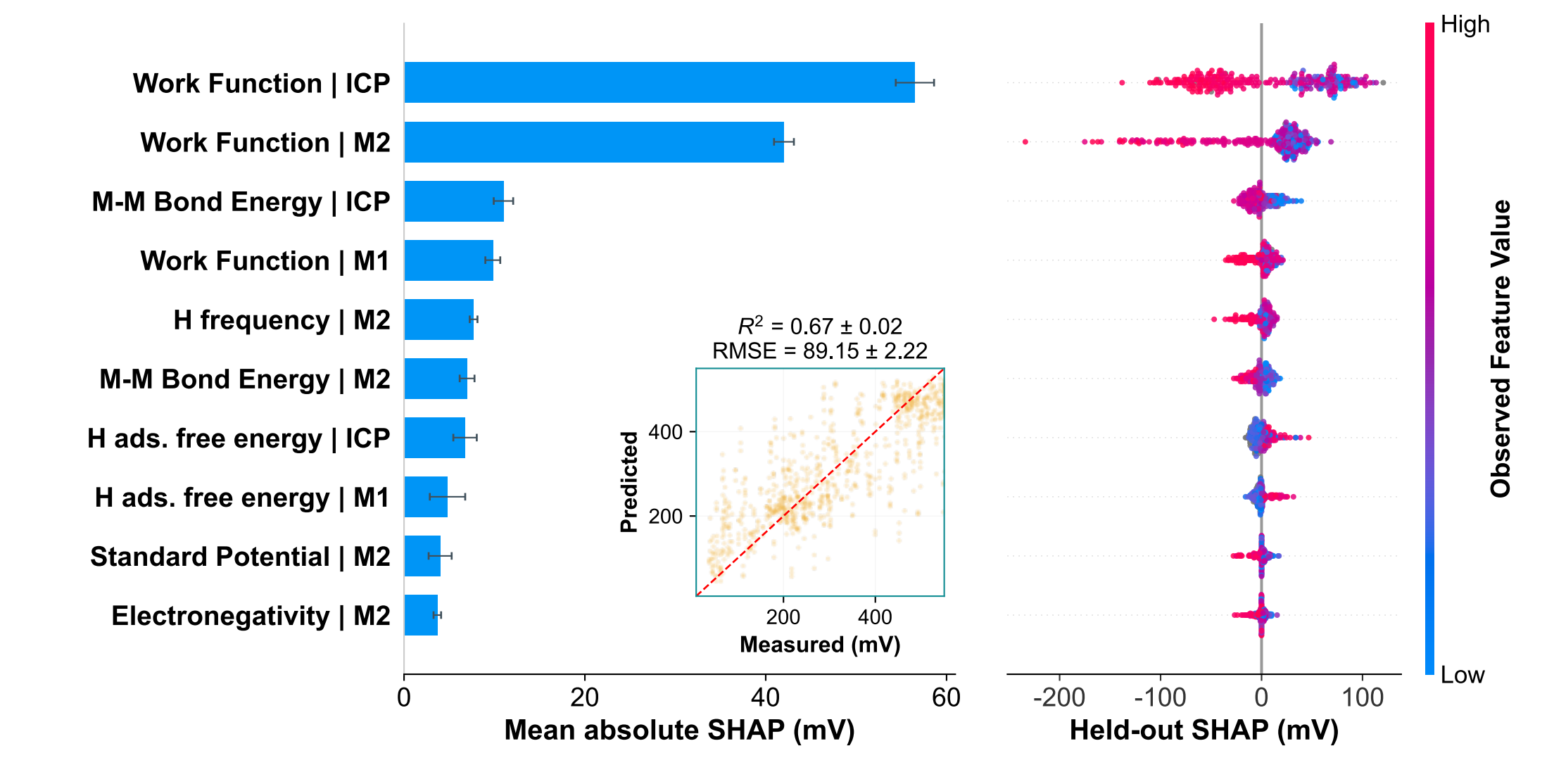
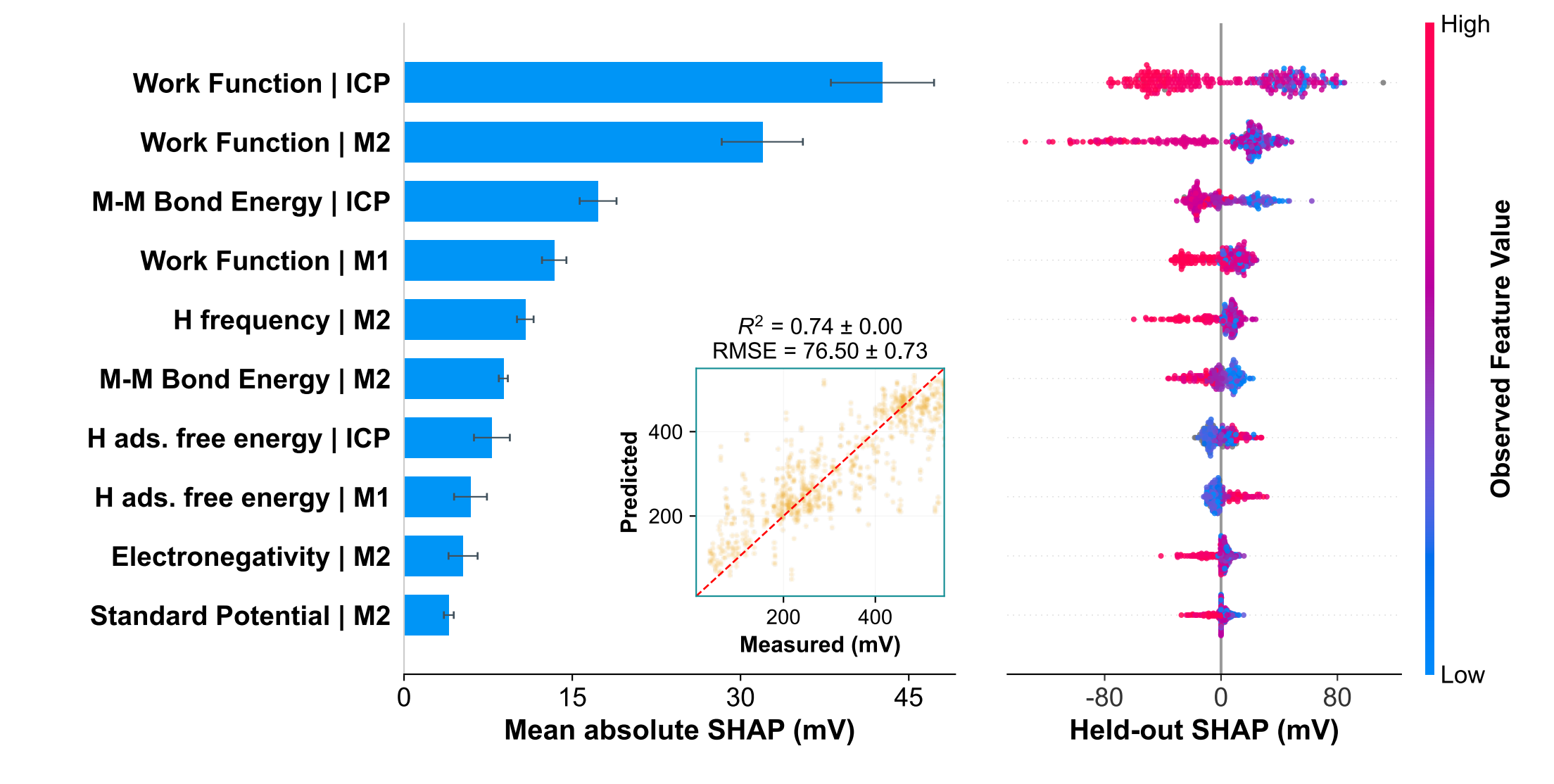

Work Function | M1
Work Function | ICP
H ads. free energy | M1
Electronegativity | M1
M-H Bond Energy | M1
PZC | M1
Work Function | M2
O-H frequency | M1
Electronegativity | ICP
H frequency | M1
0.00
0.06
0.12
0.18
Mean absolute SHAP
R^2 = 0.55 ± 0.01
RMSE = 1.12 ± 0.01
Predicted
Measured
3
6
9
-0.30
0.00
0.30
0.60
Held-out SHAP
High
Low
Observed Feature Value

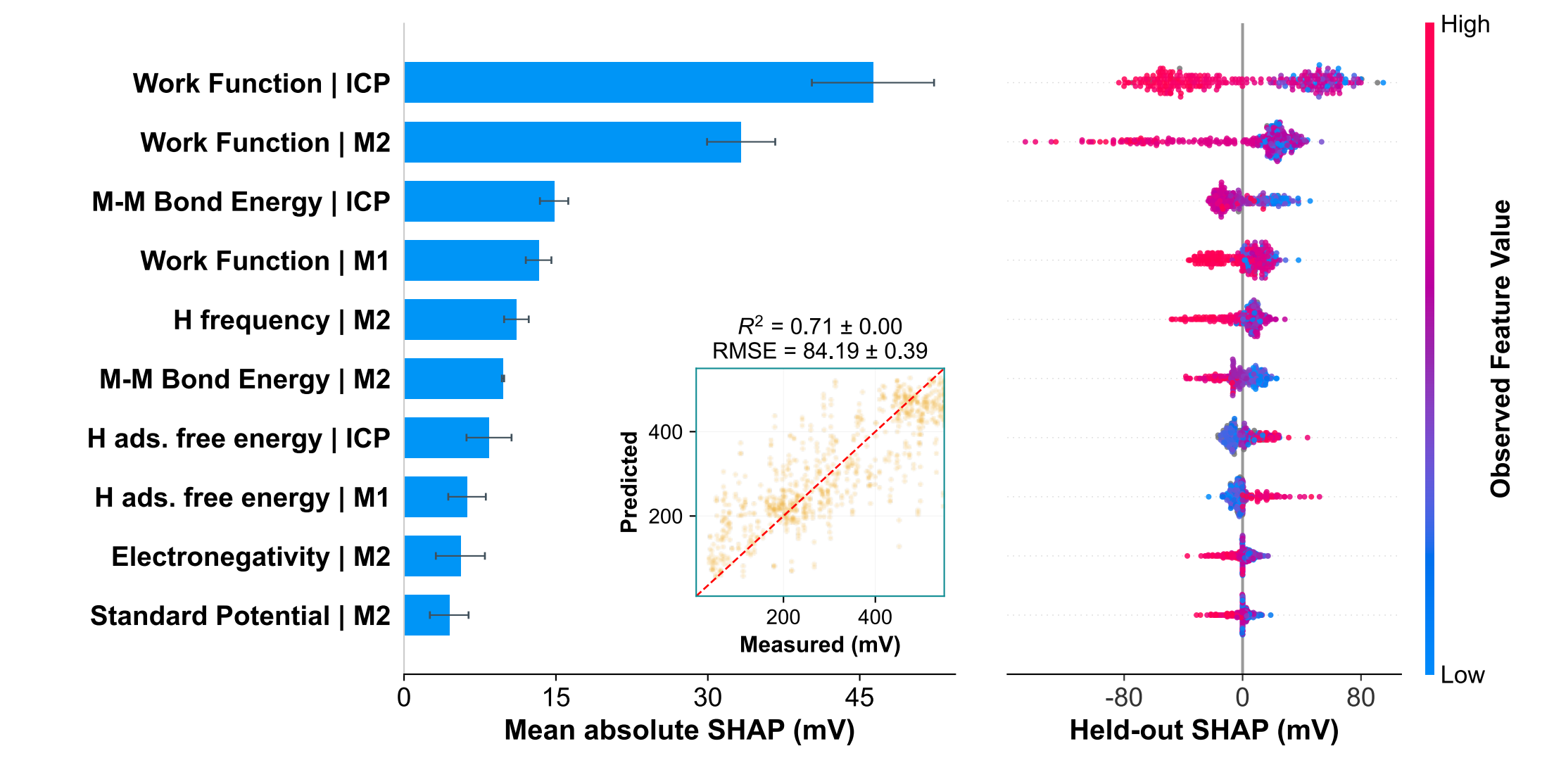

Work Function | ICP
Work Function | M1
H ads. free energy | M1
Electronegativity | M1
M-H Bond Energy | M1
PZC | M1
Electronegativity | ICP
Work Function | M2
O-H frequency | M1
H frequency | M1
0.00
0.08
0.16
0.24
Mean absolute SHAP
R^2 = 0.55 ± 0.01
RMSE = 1.11 ± 0.01
Predicted
Measured
3
6
9
-0.30
0.00
0.30
0.60
Held-out SHAP
High
Low
Observed Feature Value

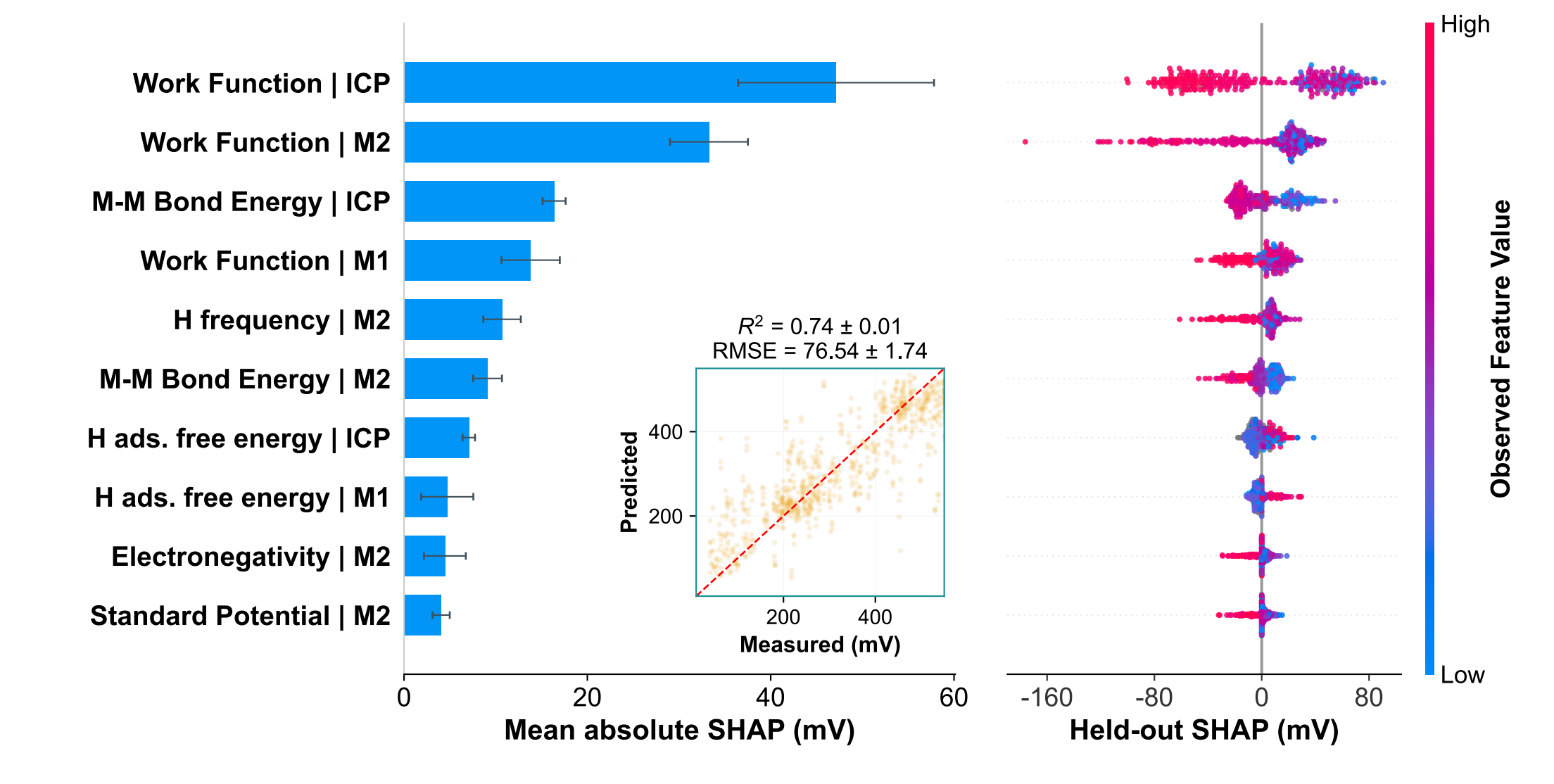

A Random, without bo
Work Function | M1
Work Function | ICP
H ads. free energy | M1
Electronegativity | M1
M-H Bond Energy | M1
PZC | M1
Work Function | M2
O-H frequency | M1
Electronegativity | ICP
H frequency | M1
0.00
0.05
0.10
0.15
Mean absolute SHAP
B Random, with bo
Work Function | M1
Work Function | ICP
H ads. free energy | M1
Work Function | M2
H frequency | M1
Electronegativity | ICP
Electronegativity | M1
OH Adsorption Energy | ICP
O-H frequency | M1
O-H frequency | ICP
0.00
0.15
0.30
0.45
Mean absolute SHAP
C Unseen pair, without bo
Work Function | ICP
Work Function | M1
H ads. free energy | M1
Electronegativity | M1
M-H Bond Energy | M1
PZC | M1
Electronegativity | ICP
Work Function | M2
O-H frequency | M1
H frequency | M1
0.00
0.05
0.10
0.15
Mean absolute SHAP
D Unseen pair, with bo
Work Function | M1
Work Function | ICP
H ads. free energy | M1
Electronegativity | M1
Work Function | M2
H frequency | M1
O-H frequency | M1
M-H Bond Energy | M1
OH Adsorption Energy | ICP
O-H frequency | ICP
0.00
0.15
0.30
Mean absolute SHAP

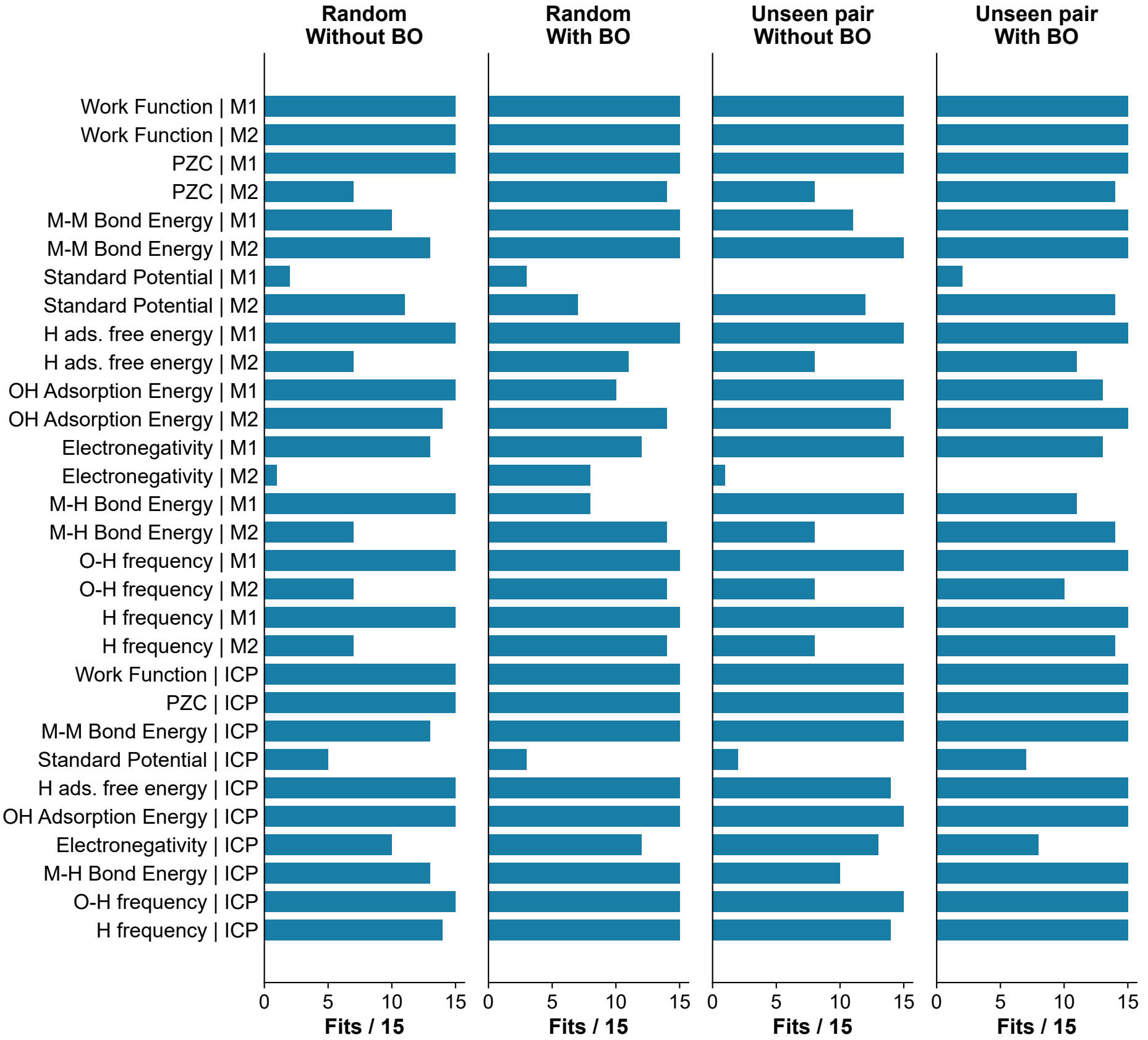

A Unshifted (30 inputs)
Work Function | M1
Work Function | ICP
H ads. free energy | M1
Electronegativity | M1
M-H Bond Energy | M1
PZC | M1
Work Function | M2
O-H frequency | M1
0.00
0.05
0.10
0.15
Mean absolute SHAP
B Unshifted + measurements (33)
Work Function | ICP
Corrected potential
Work Function | M1
H ads. free energy | M1
Electronegativity | M1
M-H Bond Energy | M1
Current near -1 V
O-H frequency | M1
0.00
0.05
0.10
0.15
Mean absolute SHAP
C Potential-shifted (30)
Work Function | ICP
Work Function | M1
H ads. free energy | M1
Electronegativity | M1
Work Function | M2
M-H Bond Energy | M1
Electronegativity | ICP
O-H frequency | M1
0.00
0.06
0.12
0.18
Mean absolute SHAP
D Shifted + measurements (33)
Work Function | ICP
Work Function | M1
H ads. free energy | M1
Electronegativity | M1
Corrected potential
M-H Bond Energy | M1
Current near -1 V
Work Function | M2
0.00
0.05
0.10
0.15
Mean absolute SHAP


Work Function | ICP
Work Function | M2
M-M Bond Energy | ICP
Work Function | M1
H frequency | M2
M-M Bond Energy | M2
H ads. free energy | ICP
H ads. free energy | M1
Standard Potential | M2
Electronegativity | M2
0
20
40
60
Mean absolute SHAP (mV)
R^2 = 0.67 ± 0.02
RMSE = 89.15 ± 2.22
Predicted
400
200
200
400
Measured (mV)
-200
-100
0
100
Held-out SHAP (mV)
High
Low
Observed Feature Value


Work Function | ICP
Work Function | M2
M-M Bond Energy | ICP
Work Function | M1
H frequency | M2
M-M Bond Energy | M2
H ads. free energy | ICP
H ads. free energy | M1
Electronegativity | M2
Standard Potential | M2
0
15
30
45
Mean absolute SHAP (mV)
R^2 = 0.74 ± 0.00
RMSE = 76.50 ± 0.73
Predicted
400
200
200
400
Measured (mV)
-80
0
80
Held-out SHAP (mV)
High
Low
Observed Feature Value


Work Function | ICP
Work Function | M2
M-M Bond Energy | ICP
Work Function | M1
H frequency | M2
M-M Bond Energy | M2
H ads. free energy | ICP
H ads. free energy | M1
Electronegativity | M2
Standard Potential | M2
0
15
30
45
Mean absolute SHAP (mV)
R^2 = 0.71 ± 0.00
RMSE = 84.19 ± 0.39
Predicted
400
200
200
400
Measured (mV)
-80
0
80
Held-out SHAP (mV)
High
Low
Observed Feature Value


Work Function | ICP
Work Function | M2
M-M Bond Energy | ICP
Work Function | M1
H frequency | M2
M-M Bond Energy | M2
H ads. free energy | ICP
H ads. free energy | M1
Electronegativity | M2
Standard Potential | M2
0
20
40
60
Mean absolute SHAP (mV)
R^2 = 0.74 ± 0.01
RMSE = 76.54 ± 1.74
Predicted
400
200
200
400
Measured (mV)
-160
-80
0
80
Held-out SHAP (mV)
High
Low
Observed Feature Value


Random
Without BO
Random
With BO
Unseen pair
Without BO
Unseen pair
With BO
Work Function | M1
Work Function | M2
PZC | M1
PZC | M2
M-M Bond Energy | M1
M-M Bond Energy | M2
Standard Potential | M1
Standard Potential | M2
H ads. free energy | M1
H ads. free energy | M2
OH Adsorption Energy | M1
OH Adsorption Energy | M2
Electronegativity | M1
Electronegativity | M2
M-H Bond Energy | M1
M-H Bond Energy | M2
O-H frequency | M1
O-H frequency | M2
H frequency | M1
H frequency | M2
Work Function | ICP
PZC | ICP
M-M Bond Energy | ICP
Standard Potential | ICP
H ads. free energy | ICP
OH Adsorption Energy | ICP
Electronegativity | ICP
M-H Bond Energy | ICP
O-H frequency | ICP
H frequency | ICP
0
5
10
15
Fits / 15